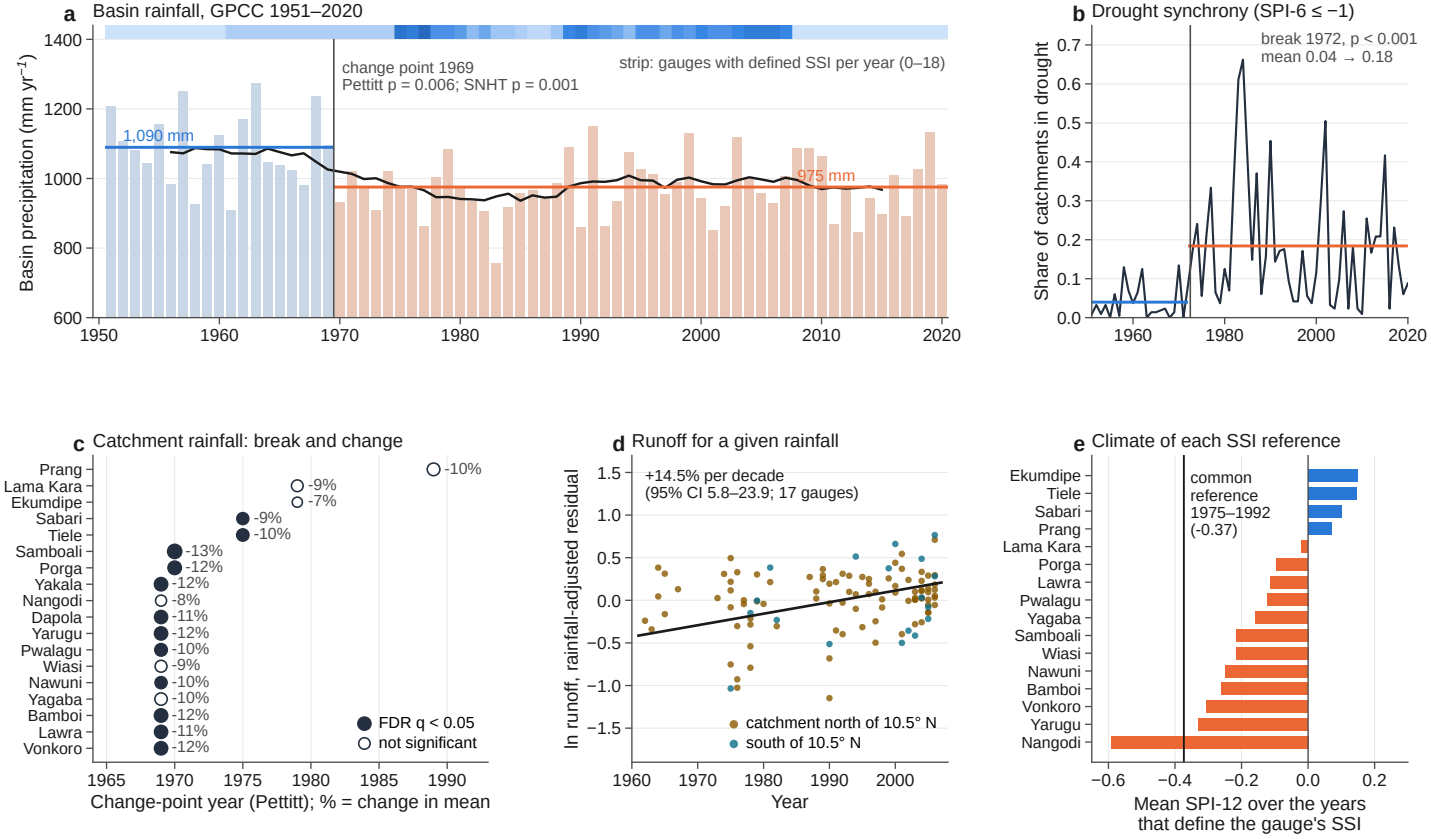
a
Basin rainfall, GPCC 1951–2020
Basin precipitation (mm yr⁻¹)
1400
1200
1000
800
600
1950
1960
1970
1980
1990
2000
2010
2020
change point 1969
Pettitt p = 0.006; SNHT p = 0.001
strip: gauges with defined SSI per year (0–18)
1,090 mm
975 mm
b
Drought synchrony (SPI-6 ≤ −1)
break 1972, p < 0.001
mean 0.04 → 0.18
Share of catchments in drought
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
1960
1980
2000
2020
c
Catchment rainfall: break and change
Prang
Lama Kara
Ekumdipe
Sabari
Tiele
Samboali
Porga
Yakala
Nangodi
Dapola
Yarugu
Pwalagu
Wiasi
Nawuni
Yagaba
Bamboi
Lawra
Vonkoro
-10%
-9%
-7%
-9%
-10%
-13%
-12%
-12%
-8%
-11%
-12%
-10%
-9%
-10%
-10%
-12%
-11%
-12%
FDR q < 0.05
not significant
1965
1970
1975
1980
1985
1990
Change-point year (Pettitt); % = change in mean
d
Runoff for a given rainfall
+14.5% per decade
(95% CI 5.8–23.9; 17 gauges)
ln runoff, rainfall-adjusted residual
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
1960
1970
1980
1990
2000
Year
catchment north of 10.5° N
south of 10.5° N
e
Climate of each SSI reference
Ekumdipe
Tiele
Sabari
Prang
Lama Kara
Porga
Lawra
Pwalagu
Yagaba
Samboali
Wiasi
Nawuni
Bamboi
Vonkoro
Yarugu
Nangodi
common
reference
1975–1992
(-0.37)
−0.6
−0.4
−0.2
0.0
0.2
Mean SPI-12 over the years
that define the gauge's SSI